2.74 3D Matrices: Transformations

Worked Solutions

1 a)

Points along the line with direction $\begin{pmatrix}1\\0\\1\end{pmatrix}$ are stretched by scale factor 2

1 b)

Points along the x-axis are stretched by scale factor 4

1 c)

The line with direction $\begin{pmatrix}1\\2\\-1\end{pmatrix}$ is invariant; This is a line of invariant points.

1 d)

The line with direction $\begin{pmatrix}0\\1\\-1\end{pmatrix}$ is invariant but points are reflected across the origin along the line.

1 e)

Points along the z-axis are stretched by scale factor 2 and points are reflected across the origin along the line.

2 a)

$$\begin{pmatrix}2&1&0\\0&3&0\\0&0&1\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\lambda\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}2&1&0\\0&3&0\\0&0&1\end{pmatrix}-\lambda\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}=\begin{pmatrix}2-\lambda&1&0\\0&3-\lambda&0\\0&0&1-\lambda\end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=(2-\lambda)(3-\lambda)(1-\lambda)$$

So $(1-\lambda)(2-\lambda)(3-\lambda)=0$

$$\therefore \lambda_1=1, \lambda_2=2 \text{ or } \lambda_3=3$$

When $\lambda_1=1$ then $\mathbf{A}-1\mathbf{I}=\begin{pmatrix}1&1&0\\0&2&0\\0&0&0\end{pmatrix}$

$$\begin{pmatrix}1&1&0\\0&2&0\\0&0&0\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=0$$

$$\Rightarrow x+y=0 \quad (1)$$
$$2y=0 \quad (2)$$

$\therefore y = -x$ But since $y = 0, x = 0$

$$\therefore \text{ when } \lambda_1 = 1, \mathbf{v}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

When $\lambda_2 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow y = 0 \quad (1)$$
$$-z = 0 \quad (3)$$

$\therefore y = 0, z = 0$

$$\therefore \text{ when } \lambda_2 = 2, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

When $\lambda_3 = 3$ then $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

$$\begin{pmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow -x + y = 0 \quad (1)$$
$$-2z = 0 \quad (3)$$

$\therefore y = x, z = 0$

$$\therefore \text{ when } \lambda_3 = 3, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\therefore \text{when } \lambda_1 = 1, \mathbf{v}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \text{when } \lambda_2 = 2, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \text{when } \lambda_3 = 3, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

2 b)

$$\det \mathbf{A} = \begin{vmatrix} 2 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 2 \times 3 \times 1 = 6$$

The transformation scales the volume by a factor of 6

2 c)

The eigenvalues and eigenvectors tell us:

- Points along the line with direction $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ are stretched by scale factor 2

- Points along the line with direction $\begin{pmatrix}1\\1\\0\end{pmatrix}$ are stretched by scale factor 3
- The z-axis is invariant; This is a line of invariant points.

3 a)

$$\begin{pmatrix}-1 & 2 & -1\\0 & 1 & 1\\0 & 0 & 2\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\lambda\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}-1 & 2 & -1\\0 & 1 & 1\\0 & 0 & 2\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{pmatrix}=\begin{pmatrix}-1-\lambda & 2 & -1\\0 & 1-\lambda & 1\\0 & 0 & 2-\lambda\end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=(-1-\lambda)(1-\lambda)(2-\lambda)$$

So $(-1-\lambda)(1-\lambda)(2-\lambda)=0$

$$\therefore \lambda_1=-1,\lambda_2=1 \text{ or } \lambda_3=2$$

When $\lambda_1=-1$ then $\mathbf{A}+1\mathbf{I}=\begin{pmatrix}0 & 2 & -1\\0 & 2 & 1\\0 & 0 & 3\end{pmatrix}$

$$\begin{pmatrix}0 & 2 & -1\\0 & 2 & 1\\0 & 0 & 3\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=0$$

$$\Rightarrow 2y-z=0 \quad (1)$$
$$2y+z=0 \quad (2)$$
$$3z=0 \quad (3)$$

$\therefore z=0\Rightarrow y=0$

$$\therefore \text{ when } \lambda_1=1,\mathbf{v}_1=\begin{pmatrix}1\\0\\0\end{pmatrix}$$

When $\lambda_2=1$ then $\mathbf{A}-1\mathbf{I}=\begin{pmatrix}-2 & 2 & -1\\0 & 0 & 1\\0 & 0 & 1\end{pmatrix}$

$$\begin{pmatrix}-2 & 2 & -1\\0 & 0 & 1\\0 & 0 & 1\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=0$$

$$\Rightarrow -2x+2y-z=0 \quad (1)$$
$$z=0 \quad (2)$$

$\therefore x=y,z=0$

$$\therefore \text{ when } \lambda_2=1,\mathbf{v}_2=\begin{pmatrix}1\\1\\0\end{pmatrix}$$

When $\lambda_3 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -3 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

$$\begin{pmatrix} -3 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow -3x + 2y - z = 0 \quad (1)$$
$$-y + z = 0 \quad (2)$$

$\therefore z = y$

Sub into (1): $-3x + y = 0 \therefore y = 3x$

$$\therefore \text{ when } \lambda_3 = 2, \mathbf{v}_3 = \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}$$

$$\therefore \text{when } \lambda_1 = -1, \mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \text{when } \lambda_2 = 1, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \text{when } \lambda_3 = 2, \mathbf{v}_3 = \begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}$$

3 b)

$$\det \mathbf{A} = \begin{vmatrix} -1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{vmatrix} = -1 \times 1 \times 2 = -2$$

The transformation scales the volume by a factor of 2 and orientation is reversed.

3 c)

The eigenvalues and eigenvectors tell us:

- Points along the x-axis are reflected across the origin along the axis.
- The line with direction $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is invariant; This is a line of invariant points.
- Points along the line with direction $\begin{pmatrix} 1 \\ 3 \\ 3 \end{pmatrix}$ are stretched by scale factor 2

4 a)

$$\begin{pmatrix} 3 & -4 & -2 \\ 0 & -1 & 0 \\ 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 3 & -4 & -2 \\ 0 & -1 & 0 \\ 0 & 2 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3-\lambda & -4 & -2 \\ 0 & -1-\lambda & 0 \\ 0 & 2 & 1-\lambda \end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I}) = (3-\lambda)(-1-\lambda)(1-\lambda)$$

So $(3-\lambda)(-1-\lambda)(1-\lambda) = 0$

$$\therefore \lambda_1 = -1, \lambda_2 = 1 \text{ or } \lambda_3 = 3$$

When $\lambda_1 = -1$ then $\mathbf{A} + 1\mathbf{I} = \begin{pmatrix} 4 & -4 & -2 \\ 0 & 0 & 0 \\ 0 & 2 & 2 \end{pmatrix}$

$$\begin{pmatrix} 4 & -4 & -2 \\ 0 & 0 & 0 \\ 0 & 2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow 4x - 4y - 2z = 0 \quad (1)$$

$$2y + 2z = 0 \quad (3)$$

$\therefore y = -z$

Sub into (1): $4x + 2z = 0 \Rightarrow z = -2x$

$$\therefore \text{ when } \lambda_1 = -1, \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$$

When $\lambda_2 = 1$ then $\mathbf{A} - 1\mathbf{I} = \begin{pmatrix} 2 & -4 & -2 \\ 0 & -2 & 0 \\ 0 & 2 & 0 \end{pmatrix}$

$$\begin{pmatrix} 2 & -4 & -2 \\ 0 & -2 & 0 \\ 0 & 2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow 2x - 4y - 2z = 0 \quad (1)$$

$$-2y = 0 \quad (2)$$

$\therefore y = 0, x = z$

$$\therefore \text{ when } \lambda_2 = 1, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

When $\lambda_3 = 3$ then $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} 0 & -4 & -2 \\ 0 & -4 & 0 \\ 0 & 2 & -2 \end{pmatrix}$

$$\begin{pmatrix} 0 & -4 & -2 \\ 0 & -4 & 0 \\ 0 & 2 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow -4y - 2z = 0 \quad (1)$$

$$-4y = 0 \quad (2)$$

$$2y - 2z = 0 \quad (3)$$

$\therefore y = 0$ and $z = y$ so $z = 0$

$$\therefore \text{ when } \lambda_3 = 3, \mathbf{v}_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\therefore \text{when } \lambda_1 = -1, \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}, \quad \text{when } \lambda_2 = 1, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \text{when } \lambda_3 = 3, \mathbf{v}_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

4 b)

$$\det \mathbf{A} = \begin{vmatrix} 3 & -4 & -2 \\ 0 & -1 & 0 \\ 0 & 2 & 1 \end{vmatrix} = 3 \times -1 \times 1 = -3$$

The transformation scales the volume by a factor of 3 and orientation is reversed.

4 c)

The eigenvalues and eigenvectors tell us:

- The line with direction $\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ is invariant but points are reflected across the origin along the line.
- The line with direction $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is invariant; This is a line of invariant points.
- Points along the x-axis are stretched by scale factor 3

5 a)

Since $-5 \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix} = \begin{pmatrix} -15 \\ -20 \\ 5 \end{pmatrix}$ then $\begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix}$ is an eigenvector.

5 b)

$\lambda = -5$

6 a)

- The line with direction $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ are stretched by scale factor 2^n
- The x-axis is invariant; points on this line remain unchanged.
- Points along the z-axis are stretched by scale factor $(-3)^n$ (3^n with a reflection if n is odd)

6 b)

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \mathbf{D} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{pmatrix} \text{ and } \mathbf{P}^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & -5 & -3 \end{pmatrix}$$

$$\therefore \mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & -5 & -3 \end{pmatrix}$$

6 c)

$$\mathbf{A}^n = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{pmatrix}^n \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$
$$= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2^n & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & (-3)^n \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$
$$= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 2^n & 0 \\ 1 & 0 & 0 \\ 0 & (-3)^n & (-3)^n \end{pmatrix}$$

$$\therefore \mathbf{A}^n = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2^n & 0 \\ 0 & -2^n + (-3)^n & (-3)^n \end{pmatrix}$$

7 a)

$$\begin{pmatrix} p & q & 0 \\ q & p & 0 \\ 0 & 0 & r \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} p & q & 0 \\ q & p & 0 \\ 0 & 0 & r \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} p-\lambda & q & 0 \\ q & p-\lambda & 0 \\ 0 & 0 & r-\lambda \end{pmatrix}$$

$$\begin{aligned} \det(\mathbf{A} - \lambda\mathbf{I}) &= (p-\lambda)(p-\lambda)(r-\lambda) - q(q)(r-\lambda) \\ &= (r-\lambda)((p-\lambda)(p-\lambda) - q(q)) \\ &= (r-\lambda)(p^2 - 2p\lambda + \lambda^2 - q^2) \\ &= (r-\lambda)(\lambda - (p+q))(\lambda - (p-q)) \end{aligned}$$

So $(r-\lambda)(\lambda-(p+q))(\lambda-(p-q)) = 0$

$$\therefore \lambda_1 = p-q, \lambda_2 = p+q \text{ or } \lambda_3 = r$$

When $\lambda_1 = p-q$ then $\mathbf{A} - (p-q)\mathbf{I} = \begin{pmatrix} q & q & 0 \\ q & q & 0 \\ 0 & 0 & r-p+q \end{pmatrix}$

$$\begin{pmatrix} q & q & 0 \\ q & q & 0 \\ 0 & 0 & r-p+q \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow qx + qy = 0 \quad (1)$$
$$(r - p + q)z = 0 \quad (3)$$

$\therefore y = -x$ and $z = 0$

$$\therefore \text{ when } \lambda_1 = p - q, \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

When $\lambda_2 = p + q$ then $\mathbf{A} - (p+q)\mathbf{I} = \begin{pmatrix} -q & q & 0 \\ q & -q & 0 \\ 0 & 0 & r - p - q \end{pmatrix}$

$$\begin{pmatrix} -q & q & 0 \\ q & -q & 0 \\ 0 & 0 & r - p - q \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow -qx + qy = 0 \quad (1)$$
$$qx - qy = 0 \quad (2)$$
$$(r - p - q)z = 0 \quad (3)$$

$\therefore z = 0, y = x$

$$\therefore \text{ when } \lambda_2 = p + q, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

When $\lambda_3 = r$ then $\mathbf{A} - (r)\mathbf{I} = \begin{pmatrix} p - r & q & 0 \\ q & p - r & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$$\begin{pmatrix} p - r & q & 0 \\ q & p - r & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow (p - r)x + qy = 0 \quad (1)$$
$$qx + (p - r)y = 0 \quad (2)$$

From (2): $y = \frac{q}{r-p}x$

Sub into (1): $(p - r)x + q\left(\frac{q}{r-p}x\right) = 0$

so for distinct eigenvalues $(r \neq p \pm q), x = y = 0$

$$\therefore \text{ when } \lambda_3 = r, \mathbf{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\therefore \text{when } \lambda_1 = p - q, \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \text{when } \lambda_2 = p + q, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \text{when } \lambda_3 = r, \mathbf{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

7 b)

$\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = 1 - 1 + 0 = 0$ so $\mathbf{v}_1$ and $\mathbf{v}_2$ are perpendicular.

$\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0 - 0 + 0 = 0$ so $\mathbf{v}_1$ and $\mathbf{v}_3$ are perpendicular.

$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0 + 0 + 0 = 0$ so $\mathbf{v}_2$ and $\mathbf{v}_3$ are perpendicular.

7 c)

The eigenvalues and eigenvectors tell us:

- Points along the line with direction $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$ are stretched by scale factor $p - q$
- Points along the line with direction $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ are stretched by scale factor $p + q$
- Points along the z-axis are stretched by scale factor r

8 a)

$$\begin{pmatrix} 1-k & k & k \\ -2 & k+1 & 2 \\ 2-2k & k & 2k-1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned} \mathbf{A} - \lambda\mathbf{I} &= \begin{pmatrix} 1-k & k & k \\ -2 & k+1 & 2 \\ 2-2k & k & 2k-1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1-k-\lambda & k & k \\ -2 & k+1-\lambda & 2 \\ 2-2k & k & 2k-1-\lambda \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \det(\mathbf{A} - \lambda\mathbf{I}) &= \begin{vmatrix} 1-k-\lambda & k & k \\ -2 & k+1-\lambda & 2 \\ 2-2k & k & 2k-1-\lambda \end{vmatrix} \\ &= \begin{vmatrix} 1-k-\lambda & k & k \\ -2 & k+1-\lambda & 2 \\ 1-k+\lambda & 0 & k-1-\lambda \end{vmatrix} \\ &= \begin{vmatrix} 1-k-\lambda & k & 1-\lambda \\ -2 & k+1-\lambda & 0 \\ 1-k+\lambda & 0 & 0 \end{vmatrix} \\ &= 0 - 0 + (1-\lambda)(-(k+1-\lambda)(1-k+\lambda)) \\ &= (1-\lambda)(k+1-\lambda)(k-1-\lambda) \end{aligned}$$

$$\therefore \lambda_1 = 1, \lambda_2 = k-1 \text{ or } \lambda_3 = k+1$$

8 b)

The eigenvalues and eigenvectors tell us:

- The line with direction $\begin{pmatrix}1\\0\\1\end{pmatrix}$ is invariant; This is a line of invariant points.
- Points along the line with direction $\begin{pmatrix}0\\1\\-1\end{pmatrix}$ are stretched by scale factor $(k-1)^3$
- Points along the line with direction $\begin{pmatrix}1\\1\\1\end{pmatrix}$ are stretched by scale factor $(k+1)^3$